

Happiness is not Mutual

Exploring Differences in the Perception of Well-being
with Words-association Network

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Highlights

This study introduces word-association network and analysis framework to explore the components, structure, and demographic differences in the perception of well-being.

The proposed method not only aligns well with previous studies' implications but also deepens our understanding of the structure by uncovering new components in well-being and examining the connections between them.

What is well-being?

Complexity in components and constitutes of well-being

Hedonic orientation

- Hedonic orientation defines well-being in terms of pleasure attainment and pain avoidance
 - **Enjoyment and pleasure**
 - Enjoying hobbies, recreational activities, or spending time with loved ones.
 - **Physical comfort**
 - Being free from chronic pain, having a comfortable living environment, and enjoying good health.
 - **Presence of positive affect**
 - Regularly feeling happy, excited, and relaxed.

Eudaimonic orientation

- Eudaimonic orientation focuses on meaning and self-realization, defining well-being in terms of the degree to which a person is fully functioning.
 - **Purpose in Life**
 - Feeling a sense of purpose in one's career and personal relationships.
 - **Personal Growth**
 - Actively seeking out to enhance one's knowledge and achieve desired outcomes.
 - **Contribution to the broader community and society.**
 - Engaging in activities that benefit others, and feeling a sense of belonging and contribution to the welfare of society.

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How perception of well-being vary?

Demographic differences in the perception of well-being

Gender

- Females' well-being is more likely to highlight the importance of **interpersonal relationships** (Karasawa et al., 2011; Matud et al., 2019) and **emotional experiences** (van der Meer, 2014).
- **Socio-economic factors**, such as employment, often have a greater impact on males' well-being (van der Meer, 2014).

Age

- **U-shaped curve** that well-being has its lowest point in middle age (Steptoe et al., 2015).
- As individuals age, the components that constitute their well-being evolve.
 - The importance of **leisure** in well-being tends to decline with age (Lelkes, 2007)
 - **Health** emerges as a more dominant factor in the well-being of older adults (Margolis and Myrskylä, 2013)

Education/Income

- Education/Income provides individuals with **resources** such as money, power, and prestige, which can help them achieve a lifestyle associated with higher well-being (Link and Phelan, 1995).
- **Intrinsic benefits** of education that enhance an individual's competencies (Becker, 1994).
- **Symbolic value** of income to enhance status, and the resulting sense of superiority and security is often associated with well-being (Frey and Stutzer, 2002).

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How to measure well-being?

"Traditional" measurement of well-being

Measure subjective well-being with self-report survey

- Employ several representative questions to evaluate to what extent they are satisfied or dissatisfied with their life (Fordyce, 1988)
 - "Overall, how satisfied are you with your life?"
- **Limitations:** Hard to capture the full spectrum of well-being.
 - Pre-defined measures might overlook specific aspects of well-being.
 - Constructs and components of well-being can vary significantly across time and countries, making it challenging to establish appropriate measurements for specific contexts.

Open-ended questions

- Use open-ended questions like "What is happiness?"
- Identify the entity related to well-being through human encoding.
- **Limitations**
 - Hard to capture the relationships among entities related to well-being.
 - Subjective bias and inefficiency of human encoding.

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Overview of Our Study

Promblem

- A comprehensive understanding of well-being therefore requires measurement that
 - capture multifaceted constructs of well-being
 - specify the demographic differences in the perception of well-being
 - can be efficiently implemented for investigations on different countries

Proposed Method

- Employs a novel approach that combines associative questionnaires with network analysis to investigate the concept of well-being.
- Provide more insights into the demographic differences from the perspective of components and semantic meanings of well-being.

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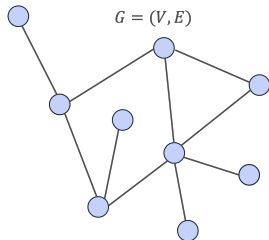
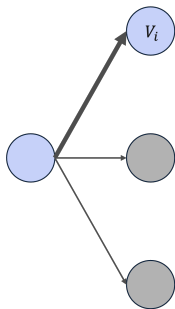
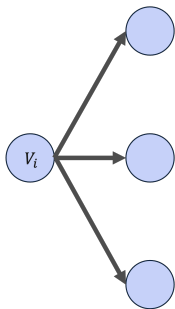
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Words-association Network

Snowballing word association exercise and construction of network

- Participants were instructed to initially list five words linked to "well-being", followed by three additional words connected to each of the initially mentioned words.



v_i : Proposed word
 E_{ij} : The number of times
word v_i and word v_j is associated

Words-association Network

Conduct survey

- Utilizing an online survey conducted in Japan, we engaged participants ranging from 20 to 79 years old and collected a total of 2,507 responses.
 - Participants were instructed to provide different words to the initial answer and to avoid responding in full sentences.

Data Processing

- Remove invalid responses, such as “no idea”, “don’t know”, or “none”.
- Select words that are mentioned at least three times.

Network Construction

- Construct an **undirected word association network** based on the data gathered from the responses.
 - Nodes represent the **words mentioned in the participants’ responses**.
 - Edges imply the **associative connections between pairs of words**.
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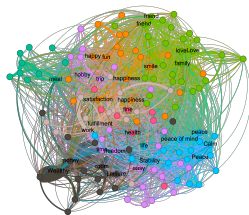
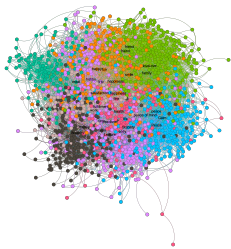
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Words-association Network

Using community detection to specify the components of well-being



- The Louvain algorithm is applied to identify communities within the word-association network.
- Detected communities are interpreted as **distinct components of well-being**.
 - Words grouped within the same community are assumed to share close semantic and contextual relationships.

- Identify the components of well-being based on key nodes in each community.
- Eight key components of well-being, including
 - “Health”
 - “Achievement”
 - “Pleasure”
 - “Family/Friends”
 - “Peacefulness”
 - “Free/Leisure”
 - “Wealth”
 - “Food”

Words-association Network

Key words contained in the respective components of well-being network.

Community	Key Words (in English)
Food	Meals, Fullness, Food, Eating, Delicious
Peacefulness	Peace, Stability, Serenity, Tranquility, Safety
Free/Leisure	Freedom, Hobby, Sleep, Travel, Time
Health	Health, Fine, Family's health, Comfort, Personal health
Wealth	Wealth, Affluence, Wealthy, Financial Freedom, Income
Pleasure	Joyful, Smiles, Happiness, Contentment, Laughter
Family/Friends	Family, Friends, Children, Love, Companions
Achievement	Work, Fulfillment, Hope, Dreams, Success

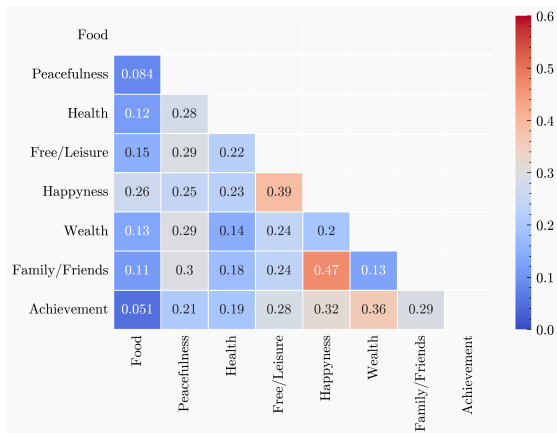
Analysis on the constructs of well-being

Inter-community connectivity of word association network

Inter-community connections

$$\frac{\text{total edge weights between two communities}}{\text{total edge weights within each community}}$$

- Inter-community connectivity informed the relationship among components.
- A higher value indicates a closer relationship.
 - “Pleasure” shows relatively high connectivity with components “Free/Leisure” and “Family and Friends”
 - “Pleasure” (0.301) have the highest average inter-community connectivity among components.



Analysis on the constructs of well-being

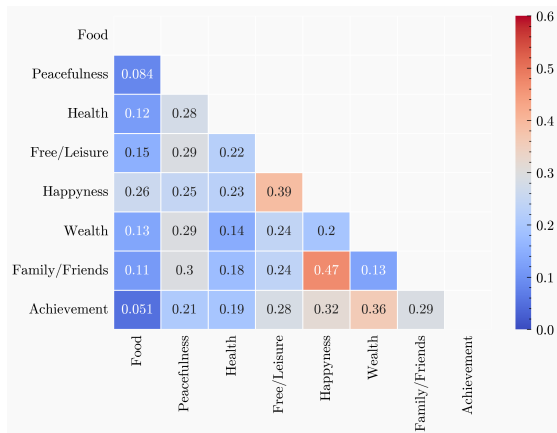
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Analysis on demographic differences

Demographic differences in each component of well-being

- Most proposed words can be classified into specific components of well-being.
- Quantify the number of distinct well-being components referenced by each participant.

$$W_i = \beta_0 + \beta_1 X_j + \epsilon$$

- W_i denotes the number of proposed words related to the component (community) of well-being i
- X_j refers to the demographic characteristics of participant j
 - Gender: 1 indicates male and 0 indicates female.
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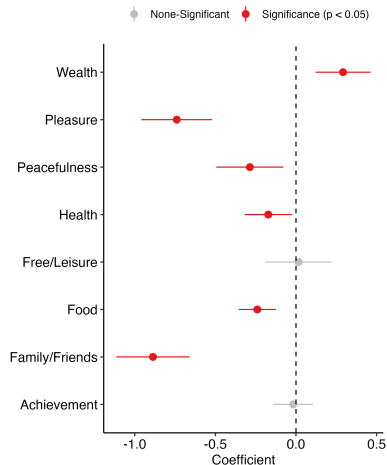
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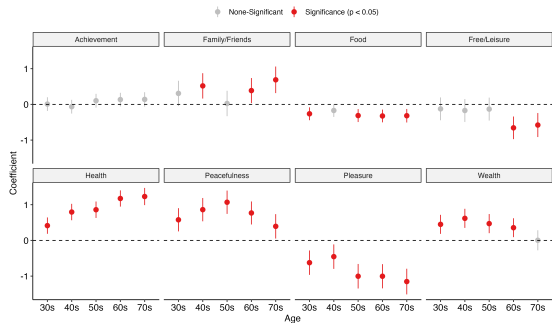
Difference in the perception of well-being between males and females

- Compared to males, females place greater value on “Family and Friends” and “Pleasure” as key components of well-being, highlighting the significant role that **social networks and emotional states play in females’ lives**.
 - Previous studies have suggested that females are typically more sensitive to psychological and affect issues and are more inclined to seek emotional support from their close persons (Fuhrer et al., 1999; Bedrov and Gable, 2022).
- **Males are more likely to consider “Wealth” an important component of well-being**
 - Societal expectations and self-identity that demand economic resources from males (Zyphur et al., 2015; Crowley, 1998; Lee et al., 2022)



Analysis on demographic differences

Difference in the perception of well-being among different age groups



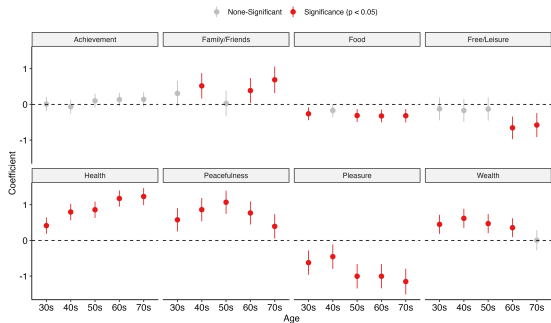
- Many factors that inhibit well-being exist in the life experiences of the middle-aged period.

- U-shaped trajectory throughout life (Step toe et al., 2015)

- Middle-aged individuals, those in their 30s to 50s, are more inclined to associate well-being with “Wealth” compared to people in their 20s and 70s.
 - Sacrifice current well-being during middle age (Step toe et al., 2015)
- With age increased, people increasingly emphasize the importance of “Health”.
 - The increasing prominence of health issues may lead to decreased well-being among middle-aged individuals.
- Middle-aged individuals still have a significant need for “Freedom/Leisure”.
 - Work responsibilities and economic pressures often prevent these needs from being fully realized.

Analysis on demographic differences

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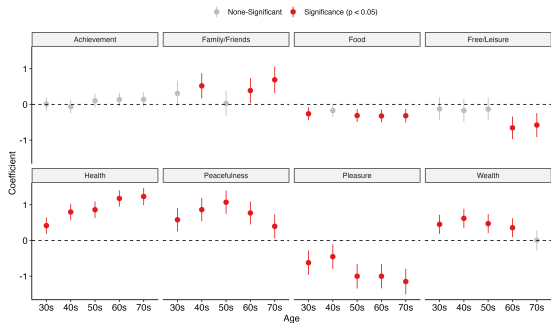
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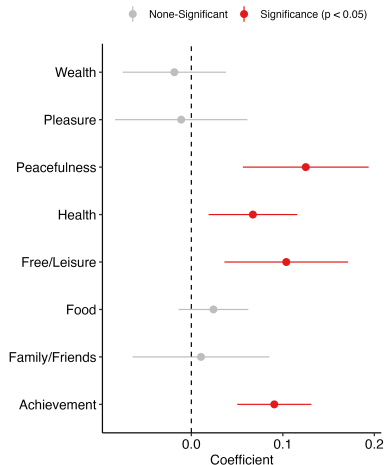
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Analysis on demographic differences

How perception of well-being related to educational attainment

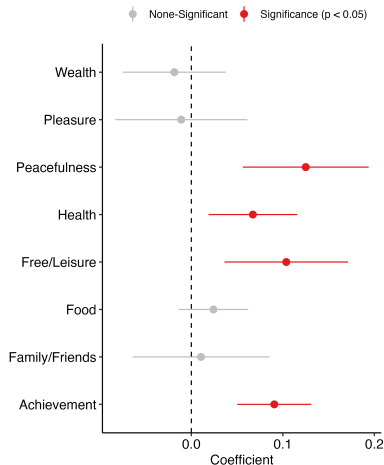
- Individuals with higher educational attainment usually place a greater emphasis on "Health" as a crucial component of well-being
 - The emphasis on health issues could result in varying health outcomes among individuals with different levels of educational attainment (Zajacova and Lawrence, 2018; Hui, 2022; Raghupathi and Raghupathi, 2020)
- Individuals with higher educational attainment tend to associate "Freedom/Leisure" and "Peacefulness" with well-being.
 - Education enables access to resources that enhance well-being (Link and Phelan, 1995)
- Individuals with higher educational attainment tend to associate "Achievement" with well-being
 - Education can enhance eudaimonic well-being (Miquelon and Vallerand, 2006)



Analysis on demographic differences

How perception of well-being related to educational attainment

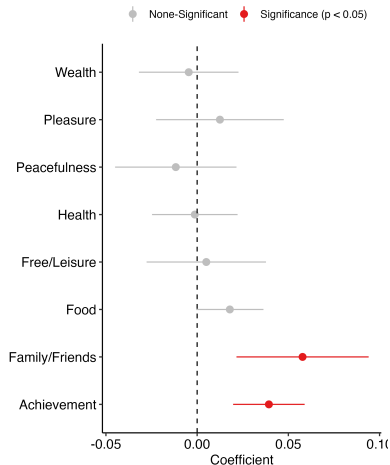
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Analysis on demographic differences

How perception of well-being related to Income

- Individuals with a higher income are more likely to emphasize “Achievement” as an important component of well-being.
 - Similar to educational attainment
 - Income can enhance **eudaimonic well-being**.
 - The recognition and respect they receive from others for their achievements can significantly boost their self-esteem and overall sense of well-being.



Analysis on demographic differences

Demographic differences from semantic perspectives

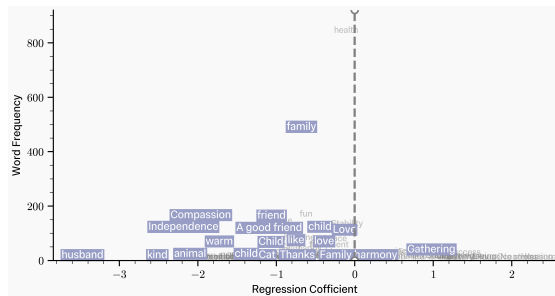
- **Motivation:** Even within the context of identical components, the emphasis placed on specific aspects varies among different demographic groups, and this nuance is typically reflected in the semantic meanings of proposed words.
- Estimate the effect of demographic characteristics on proposed words

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Analysis on demographic differences

Semantic difference in the perception of Family/Friends between males and females



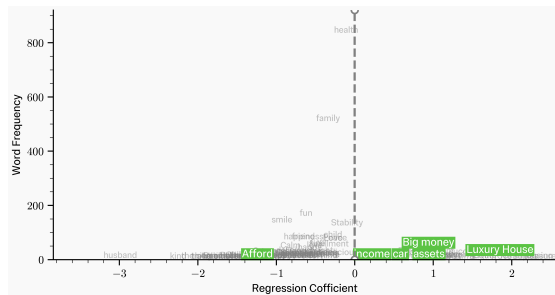
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 - The original answer is in Japanese, translated by Google Translation

Females tend to place a greater emphasis on interpersonal connections than males, while it is unclear who exactly is included in these supportive connections.

- “family”, “friends” and “love” are more likely to be proposed by females.
- “animals” and “cat”
 - Animals are increasingly being recognized as a source of emotional support and considered by many owners as friends and affectionate family members (Meehan et al., 2017).

Analysis on demographic differences

Semantic difference in the perception of Wealth between males and females



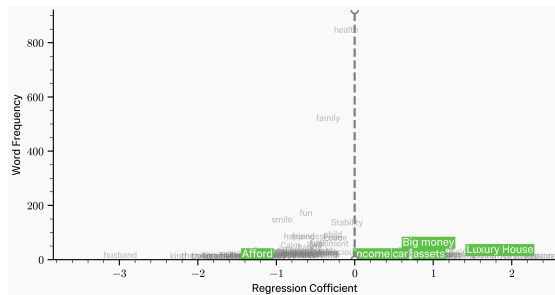
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Although previous analyses have indicated gender differences on a general level, further semantic analysis reveals more detailed underlying differences.

- Males are more likely to place greater importance on material possessions that display their economic status, such as "houses" and "cars".
- This tendency reflects societal expectations for males to demonstrate success and stability through visible symbols of wealth and achievement.

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- **Offer a analysis framework to more comprehensively understand well-being through exploratory analysis.**
 - Measurement with pre-defined categories may be incomplete, especially since the components well-being may be context-dependent.
 - Exploratory analysis can more efficiently deconstruct complex concepts and uncover aspects that may be overlooked
- **Enables a more detailed investigation of demographic differences.**
 - The structure of word-association network provides a macroscopic view of how different components are interconnected across demographic groups.
 - Semantic details offer a closer insights into the meanings and nuances of words within these connections.

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Future directions

- The proposed method and analysis framework possess broad universality and applicability, and thus can be effectively leveraged in deconstructing and analyzing various complex concepts.
 - Simplify the process for researchers by only requiring them to guide participants to complete word association tasks during data collection.
 - Avoids the extensive efforts typically required to pre-define, delineate, and analyze complex concepts, streamlining the research process significantly
- We have used the same method to investigate ill-being.

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